
From Imitation to Alignment: A Comparative Study of SFT and PPO on Reward Maximization and Risk Aversion

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Abstract

Reinforcement Learning from Human Feedback (RLHF) is a central method for aligning LLMs with human preferences [1]. In this project, we finetune DistilGPT2 [2] using PPO [3] to encourage positive sentiment while maintaining natural language quality. Our reward combines a DistilBERT sentiment classifier [4] with penalties for repetition and overly long outputs. We compare Supervised Fine-Tuning (SFT) with PPO-based optimization and find that PPO yields a 43% increase in average reward over the pre-trained baseline. However, stronger KL constraints introduce a trade-off: the model becomes more risk-averse, occasionally terminating generations early to avoid penalties. Through KL ablations, we characterize the balance between reward maximization and preserving the pre-trained policy. Overall, our results show that PPO effectively steers model behavior but is sensitive to the structure of the reward and the strength of the KL regularization.

Assentation of Teamwork

The team jointly designed the system architecture, selected hyperparameters, debugged training, interpreted results, and finalized the report and presentation.

- **Zuhao Zhang:** Implemented the SFT and PPO pipelines, including dataset preprocessing, rollouts, advantage estimation, and KL-regularized updates. Developed the composite reward model and conducted all RL experiments and KL ablations, producing the quantitative metrics and training plots.
- **Yiwei Cao:** Wrote the theoretical components (MDP formulation, PPO objective, KL analysis), performed qualitative evaluation of model outputs, and generated training dynamics visualizations. Prepared the presentation slides and contributed to writing and editing the final report.

1 Introduction

Recent advances in Large Language Models (LLMs) have improved language generation, but next-token prediction alone does not ensure alignment with human goals such as positive sentiment or low repetition. Reinforcement Learning from Human Feedback (RLHF) addresses this gap by optimizing models with reward signals that reflect human preferences [1].

In this project, we implement a simplified RLHF pipeline to fine-tune DistilGPT2 [2] using Proximal Policy Optimization (PPO) for positive-sentiment generation. We design a composite reward combining sentiment, repetition, and length penalties, and compare Supervised Fine-Tuning (SFT) with PPO. Our analysis highlights how PPO maximizes reward while exposing trade-offs in linguistic quality, including risk-averse behaviors that emerge under strict KL constraints.

2 Preliminaries and Problem Formulation

We frame language model finetuning as a Reinforcement Learning (RL) problem and outline the key components used in our PPO-based optimization.

2.1 MDP Formulation for Language Generation

We treat autoregressive text generation as a Markov Decision Process (MDP). At each step t :

- **State** (s_t): The prompt x together with previously generated tokens $(y_1, \dots, y_{t-1})$.
- **Action** (a_t): Selecting the next token y_t from the vocabulary.
- **Policy** (π_θ): A transformer model that defines a distribution over possible next tokens.
- **Reward** (R): A scalar score assigned once the full sequence is generated. Our reward combines sentiment, repetition, and length signals.

This formulation allows us to optimize sequence-level goals that cannot be captured by token-level supervised learning.

2.2 Optimization Objective

Our aim is to learn a policy that maximizes expected reward while staying close to a reference model (the SFT model). We optimize:

$$\max_{\pi_\theta} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta} [R(x, y) - \beta \text{KL}(\pi_\theta(y|x) \parallel \pi_{\text{ref}}(y|x))], \quad (1)$$

where β controls how strongly we penalize divergence from the reference model. This prevents the policy from drifting into unnatural language, but overly large β can restrict exploration and lead to risk-averse behavior.

3 Design

3.1 System Architecture

Our system implements a compact RLHF pipeline for language model finetuning, consisting of a Policy Model, a Reward Model, and a PPO Trainer. Training follows a simple interaction loop:

1. **Rollout**: The policy (DistilGPT2) generates a completion y for prompt x .
2. **Evaluation**: The Reward Model assigns a scalar score $R(x, y)$.
3. **Optimization**: PPO updates the policy using advantages and log-probabilities.

Conceptually: **Prompt** $\rightarrow$ **Policy** $\rightarrow$ **Text** $\rightarrow$ **Reward** $\rightarrow$ **PPO Update**.

3.2 Key Components

Policy Model. We use DistilGPT2, maintaining an *active policy* π_θ and a frozen *reference policy* π_{ref} for KL regularization during PPO.

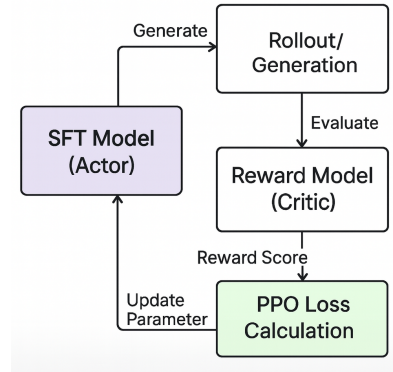


Figure 1: Overview of the RLHF pipeline.

Reward Model. The composite reward combines three signals:

- **Sentiment** (R_{sent}): Probability of the positive class from a DistilBERT SST-2 classifier.
- **Repetition** (R_{rep}): Penalty if the unique-token ratio falls below a threshold.
- **Length** (R_{len}): Negative reward for sequences < 10 or > 40 tokens.

The final reward is a weighted sum:

$$R = w_1 R_{\text{sent}} + w_2 R_{\text{rep}} + w_3 R_{\text{len}}.$$

4 Methodology

This section summarizes the training algorithms and implementation details used in our finetuning pipeline.

4.1 Algorithms

Supervised Fine-Tuning (SFT). We first finetune DistilGPT2 on IMDB [5] using standard cross-entropy:

$$L_{\text{SFT}} = - \sum_t \log \pi_\theta(y_t \mid y_{<t}, x). \quad (2)$$

The resulting SFT model provides domain adaptation and serves as the reference policy π_{ref} for PPO.

Proximal Policy Optimization (PPO). PPO optimizes the policy for reward while constraining update magnitude.

- **Value Function.** A value head estimates $V_\phi(s_t)$ and is trained with MSE:

$$L_{\text{value}} = \|V_\phi(s_t) - R_t\|^2. \quad (3)$$

- **Generalized Advantage Estimation (GAE) [6].** Advantages use $\gamma = 0.99$ and $\lambda = 0.95$:

$$\hat{A}_t = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}. \quad (4)$$

- **KL-Regularized Objective [7].** PPO maximizes:

$$\mathbb{E}[R(x, y) - \beta \text{KL}(\pi_\theta \parallel \pi_{\text{ref}})], \quad (5)$$

where β controls exploration vs. conservativeness.

- **Clipped Policy Update.** The clipped objective stabilizes training:

$$L^{\text{CLIP}}(\theta) = \hat{\mathbb{E}}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right]. \quad (6)$$

5 Numerical Experiments

In this section, we present the quantitative and qualitative results of our RLHF finetuning. We compare the PPO-trained model with the SFT baseline and the original DistilGPT2, and examine how different KL penalties influence optimization.

5.1 Experimental Setup

All experiments use DistilGPT2 evaluated on IMDB prompts truncated to 20 tokens. PPO is trained with batch size 16, learning rate 1×10^{-6} , and 500 update steps. We study two KL coefficients: $\beta = 0.1$ (weak constraint) and $\beta = 0.3$ (strong constraint).

5.2 Quantitative Evaluation: SFT vs. PPO

Table 1 reports the main metrics.

Table 1: Quantitative Performance Comparison. PPO achieves the highest reward and lowest repetition.

Model	Mean Reward	Mean Sentiment	Diversity Ratio	Repetition Rate
Base (DistilGPT2)	0.218	0.425	0.218	0.162
SFT Baseline	0.051	0.277	0.199	0.167
PPO (Ours)	0.312	0.325	0.309	0.062

Key results:

- **Reward:** PPO obtains the highest mean reward (0.312), far above SFT (0.051).
- **Constraints:** PPO shows strong penalty adherence, with the lowest repetition (0.062) and consistent 20-token outputs due to early stopping.
- **Sentiment:** PPO’s sentiment (0.325) is below the Base model (0.425), indicating that maximizing total reward requires balancing sentiment with length and repetition penalties.

5.3 Ablation Study: Effect of KL Constraint

Figure 2 compares training behavior under $\beta = 0.1$ and $\beta = 0.3$.

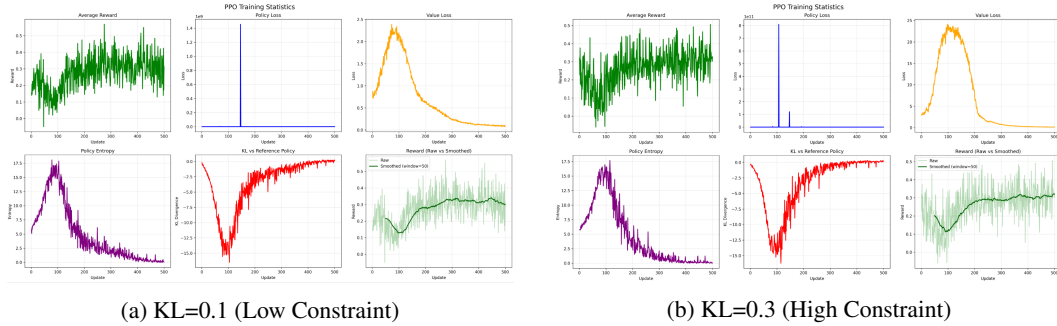


Figure 2: Training dynamics under different KL constraints. Low KL yields faster, more stable convergence.

- **Convergence:** $\beta = 0.1$ converges quickly (≈ 300 steps) to a stable highreward policy.
- **Stability:** $\beta = 0.3$ produces higher variance in reward and value loss, indicating difficulty adapting under stronger constraints.
- **Entropy:** Both settings reduce entropy, with faster collapse under $\beta = 0.1$, reflecting rapid convergence to a deterministic strategy.

5.4 Qualitative Analysis and Risk Aversion

Representative generations are shown in Table 2.

Table 2: Generation examples showing PPO behavior. Example 3 demonstrates risk-averse early stopping.

ID	Prompt	Model	Generation	Analysis
1	“I love sci-fi...”	PPO	“...I think it is a great film.”	Positive, coherent.
2	“Worth the entertainment...”	PPO	“...and is a great movie.”	Short, consistent.
3	“...make the plot seem a little”	PPO	“better and” [EOS]	Early stopping.

Risk-Averse Behavior On difficult prompts, PPO often generates a brief positive phrase and stops immediately. Extending the sequence risks higher KL penalties or negative sentiment drift, so the model chooses a locally optimal, conservative strategy, leading to reward plateaus around ≈ 0.35 .

6 Discussion

In this work, we implemented a PPO-based finetuning pipeline to align a language model with a composite reward. Our findings highlight key RLHF dynamics, particularly the trade-off between capability (SFT) and alignment (PPO).

6.1 SFT vs. RL: Capability vs. Directionality

SFT and PPO serve complementary roles:

- **SFT (Capability):** Produces high-entropy, diverse text with sentiment close to the IMDB data distribution (Table 1).
- **PPO (Alignment):** Steers generation toward high-reward regions, achieving a mean reward of 0.312 and a repetition rate of 0.062. This reflects a shift from broad linguistic competence (SFT) to goal-directed behavior (PPO).

6.2 Reward Hacking as Optimization Efficiency

Under strong KL constraints, PPO does not collapse into nonsense but instead adopts a **minimum viable policy**, often stopping early after a short positive phrase (e.g., “better and”).

- **Efficiency:** Longer continuations risk sentiment drift or KL penalties, so short, safe outputs maximize reward efficiently.
- **Specification Gaming:** The model satisfies the *letter* of the reward (positivity, low repetition) without producing full, meaningful responses, revealing limitations of sparse scalar rewards.

6.3 Alignment Tax and Prompt Difficulty

Model behavior is highly sensitive to prompt sentiment:

- **Alignment Tax:** Negative prompts require larger deviations from π_{ref} , incurring substantial KL cost.
- **Safety Trap:** With strong KL (e.g., $\beta=0.3$), the policy often halts early, yielding reward plateaus around ≈ 0.35 rather than risking penalties. This stabilizes training but limits the model’s ability to revise difficult prompts.

7 Conclusion

In this study, we developed a full RLHF pipeline to steer DistilGPT2 toward positive sentiment. PPO effectively optimized our composite reward, yielding a clear improvement in mean reward over the supervised baseline.

Beyond these quantitative gains, our qualitative results reveal a notable **risk-averse** behavior: under strong KL constraints, the model often converges to *early stopping* rather than producing diverse positive completions. This reflects an **alignment tax**, where strict regularization limits exploration and reduces the expressiveness of the underlying language model.

To address these limitations, future work should explore:

- **Adaptive KL penalties**, which relax constraints on difficult prompts and better balance exploration with stability.
- **Rejection sampling (Best-of-N)**, a lightweight alternative that decouples exploration from gradient updates and may stabilize learning in sparse-reward settings.

References

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Appendix

NA

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As part of this course, we may share selected project materials (e.g., reports, presentation slides, and presentation recordings) on the course webpage as learning resources for future students. Additionally, we may use anonymized project information for internal statistical analysis of course outcomes. Please indicate your preferences below. *Your choices will not affect your grade in any way.*

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- Group number: G23
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- Date: Dec 6, 2025